

TN-NTN Aggregation and Unification in 6G

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Abstract—In this paper, required features for terrestrial network (TN) and non-terrestrial network (NTN) unification in the sixth-generation (6G) wireless communication systems is discussed. Thanks to the reducing cost for satellite launching, NTN has drawn attention as a new solution to provide global coverage of mobile communications for the past few years. To meet the 6G data rate requirement, which is 10 times higher than that of 5G, while providing world-wide coverage at lost cost, enabling of three-dimensional (3D) network via TN-NTN aggregation and unification will be one of the design principle of 6G. This paper reveals key requirements for TN-NTN aggregation and unification, including possible dynamic ranges at UE side for simultaneous TN-NTN operations.

Index Terms—6G, TN-NTN aggregation, TN-NTN unification, 3D networks

I. INTRODUCTION

Thrid generation partnership project (3GPP) has specified the first set of specification supports for non-terrestrial network (NTN) based on a couple of study phases in the fifth generation (5G) new radio (NR) [1]–[3]. In this version of 3GPP specifications, only a set of simple level of interworking between TN and NTN, such as handover (HO), are available and simultaneous operations of TN and NTN from a user equipment (UE) perspective were left for future releases like the sixth generation (6G) networks [4], [5]. Since it is impossible to meet the 6G data rate requirements (e.g., 1Tbps from peak data rate perspective, world-wide service area from coverage perspective) by relying on either of TN or NTN, unification of TN and NTN would be one of the key features to realize both coverage and data rate requirements for 6G.

Figure 1 shows the expected evolution of NTN specifications for TN-NTN unification in 6G. As in Fig. 1, UEs following 5G NR NTN specifications from release 17 to 19 do not need to maintain TN link and NTN link simultaneously, which means that a UE cannot dynamically exchange the serving link from TN link to NTN link or vice versa and one link should be de-activated until the UE finalize cell re-selection procedure or HO procedure. Therefore, the following two approaches can be considered in upcoming releases for simultaneous operation of TN and NTN or dynamic switching of TN and NTN; 1) TN-NTN aggregation, and 2) TN-NTN unification. In the

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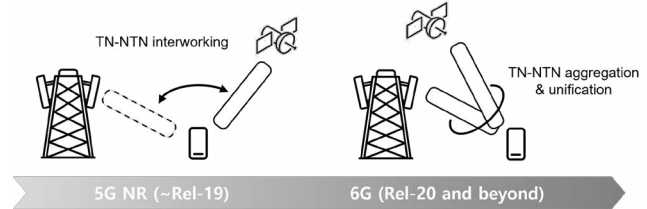


Fig. 1. Evolution towards TN-NTN unification.

following sections, we discuss details on TN-NTN aggregation and unification, which cannot be supported by the existing 5G NR specifications.

II. A BRIEF REVIEW ON 5G NTN

In this section, a brief summary on 5G NTN specification supports and their limitations is provided.¹

- **Timing and synchronization:** downlink (DL) and uplink (UL) frame timing can be aligned at the uplink time synchronization reference point (RP) existing on a certain location of feeder link between NTN payload (i.e., satellite) and NTN gateway. A base station (BS) can configure UE with common timing advance (common TA) to compensate round trip time (RTT) between NTN payload and RP. The BS further provides UE with NTN payload ephemeris information, so the UE can calculate the distance between NTN payload and itself, d_A , based on the UE location from global navigation satellite system (GNSS). The UE shall apply the UE-specific TA derived from d_A to access the NTN link. Total TA value is the sum of common TA, UE-specific TA, and legacy TA as in TN.
- **Mobility:** 5G NR provides various features for NTN mobility support. For instance, the network can provide UEs with neighbouring cell's NTN payload ephemeris as well as the serving cell's NTN payload ephemeris to assist cell re-selection of UEs. In addition, the network can broadcast TN coverage area information, so the UE can selectively perform TN measurement according to that information. For UEs in radio resource control (RRC)

¹In this paper, we highlighted a subset of NTN features for ease of understanding. See [1] and [4] for the other 5G NR NTN features.

connected mode, 5G NR NTN specifications support handover (HO), conditional handover (CHO), and satellite switch with re-synchronization features. It is noted that a UE does not need to maintain both TN and NTN links at the same time during the TN-NTN mobility procedure, which means that features like dual active protocol stack (DAPS) HO is not supported for 5G NR NTN.

- Other assumptions: 5G NR NTN specifications are clarifying that only the UEs with GNSS capability (i.e., the UEs can perform the aforementioned timing and synchronization features) are permitted to access the NTN cells. In 5G NR NTN, carrier aggregation (CA) and dual connectivity (DC) features are not supported under the same design principle that an NTN UE is not required to maintain multiple links at the same time.

III. REQUIRED FEATURES FOR TN-NTN AGGREGATION AND UNIFICATION IN 6G

In contrast with the 5G NR NTN specification supports, maintaining multiple links for an NTN UE would be the first step for TN-NTN aggregation and unification. In this section, potential issues from simultaneous operations of TN-NTN or NTN-NTN links are discussed.

Figure 2 represents possible dynamic ranges at RX side for simultaneous TN-NTN or NTN-NTN operations in S-band (CF=2GHz). In Fig. 2, each line depicts the gap between TX equivalent isotropic radiated power (EIRP) and free space pathloss (FSPL) for TN (red line, distance between gNB and UE = 250m, Tx EIRP = 67dBm as in UMi [6]), LEO-600km (cyan line, Tx EIRP = 74dBm as in study case (SC) 10 [3]), LEO-1200km (green line, Tx EIRP = 80dBm as in SC15 [3]), and GEO (blue line, Tx EIRP = 99dBm as in SC5 [3]), respectively. In 6G network, various combinations of TN-NTN or NTN-NTN links, e.g., TN-LEO links, TN-GEO links, LEO-GEO links, etc., can be considered. In such scenarios, the possible dynamic range at the receiver side can be up to 70 dB, and it may force to utilize high-end RX components, such as tens of bits analog-to-digital converter, low noise amplifier with wide dynamic range, etc. Furthermore, as shown in Fig. 2, required dynamic range may vary if one of the NTN links is served by LEO. For LEO with 600 km altitude and 30° of minimum elevation angle, variance of the gap can be more than 5 dB.

In this regard, the following features are desirable for 6G to enable TN-NTN aggregation and unification:

- Dynamic range information exchange: Given that the acceptable dynamic range of received signals for simultaneous operations of multiple links is up to UE implementation, a UE can report to network on the desired dynamic range for simultaneous TN-NTN or NTN-NTN operations. Based on such report, the network can decide and/or configure proper actual/maximum dynamic ranges for simultaneous TN-NTN or NTN-NTN operations.
- Network architecture with repeater at UE side: Asymmetric network architectures for different links can be one of solutions to mitigate the dynamic range. For instance,

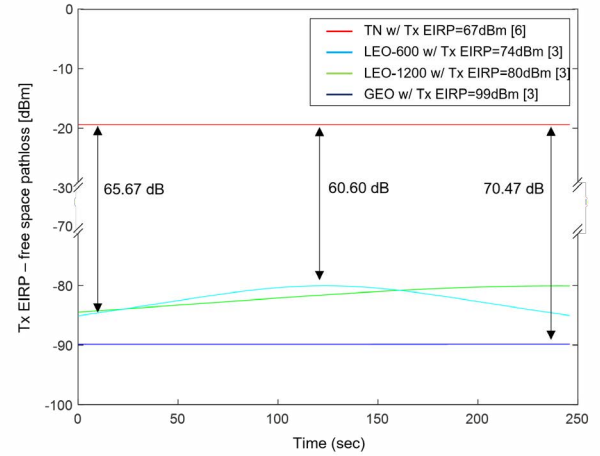


Fig. 2. Possible dynamic ranges at RX for simultaneous TN-NTN or NTN-NTN operations in S-band.

NTN link may have additional repeater node to alleviate the dynamic range at RX for simultaneous TN-NTN operations. In another example, GEO link may include additional repeater node to ease the dynamic range at RX for simultaneous GEO-LEO operations. In this option, the repeater gain/power control should be optimized to minimize the dynamic range.

- DL and UL power control: Received power of signals from two different links can be aligned to release the dynamic range at UE RX by DL power control/adjustment. Similarly, transmitted power of signals to those links can be managed by UL power control to reduce the dynamic range at UE TX.

IV. CONCLUSION

This paper investigated simultaneous operations of TN-NTN or NTN-NTN links via TN-NTN aggregation and unification to realize both high data rate requirements and wide coverage requirements in 6G. A set of key techniques, such as dynamic range information exchange, asymmetric network architectures for multiple links, DL and UL power control, etc., were discussed.

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